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## Integrable Systems — Exercise Sheet 0

*Why integrability?*

**Variant: Questions only**

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This sheet assumes only what Lecture 0 assumes: calculus, linear algebra, ordinary differential equations, and mechanics in Hamiltonian form,  $\dot{q} = \partial H / \partial p$ ,  $\dot{p} = -\partial H / \partial q$ . A quantity  $F(q, p)$  is conserved when  $\dot{F} = 0$  along every solution.

**Exercise 1 (the Hooke chain).** Take  $N$  unit masses on a line, joined by unit springs, with both ends fixed: with  $x_n$  the displacement of mass  $n$  and  $x_0 = x_{N+1} = 0$ ,

$$H = \frac{1}{2} \sum_{n=1}^N p_n^2 + \frac{1}{2} \sum_{n=0}^N (x_{n+1} - x_n)^2.$$

(i) Write Hamilton's equations, and show that  $x_n(t) = \sin(nk\pi/(N+1)) \cos \omega t$ , for  $k \in \{1, \dots, N\}$ , solves them exactly when  $\omega = \omega_k = 2 \sin(k\pi/(2(N+1)))$ , the frequencies of the lecture. (ii) Show that the vectors  $s_k = (\sin(nk\pi/(N+1)))_{n=1}^N$  are orthogonal, with  $|s_k|^2 = (N+1)/2$ . With  $Q_k = \sqrt{2/(N+1)} \sum_n \sin(nk\pi/(N+1)) x_n$  and  $P_k$  the same combination of the  $p_n$ , show that  $H = \sum_k \frac{1}{2} (P_k^2 + \omega_k^2 Q_k^2)$  and that each mode energy  $E_k = \frac{1}{2} (P_k^2 + \omega_k^2 Q_k^2)$  is conserved:  $N$  conserved quantities. (iii) For  $N = 2$ , write  $E_1$  and  $E_2$  in terms of  $x_1, x_2, p_1, p_2$  and check directly from Hamilton's equations that each is conserved. (iv) For  $N = 32$ , the chain of the lecture's figure, compute  $\omega_1$ , the period of mode 1, and how many of its periods fit into  $t = 10^4$ . Compute  $\omega_2/\omega_1$ ,  $\omega_3/\omega_1$  and  $\omega_4/\omega_1$ .

**Exercise 2 (collisions with unequal masses).** Two particles of masses  $m_1, m_2 > 0$  collide elastically on a line, particle 1 on the left, with velocities  $v_1 > v_2$  before and  $w_1, w_2$  after. (i) Solve the conservation of momentum and of energy for  $(w_1, w_2)$ . Show that the solutions are  $(w_1, w_2) = (v_1, v_2)$  and

$$w_1 = \frac{(m_1 - m_2)v_1 + 2m_2v_2}{m_1 + m_2}, \quad w_2 = \frac{(m_2 - m_1)v_2 + 2m_1v_1}{m_1 + m_2}.$$

From now on suppose the particles cannot pass through each other, so that particle 1 stays on the left. Show that this rules out the first solution and leaves the second. (ii) For the second solution, show that the centre-of-mass velocity is unchanged and the relative velocity reverses:  $w_2 - w_1 = -(v_2 - v_1)$ . (iii) For the second solution, show that  $\{w_1, w_2\} = \{v_1, v_2\}$  as sets only if  $m_1 = m_2$ . (iv) A particle of mass 1 moving at velocity 1 hits a particle of mass 3 at rest. Find the velocities after the collision, and compare with two equal masses. What does this say about a one-dimensional gas of two species?

**Exercise 3 (counting unknowns against laws).** (i) Two particles of masses  $m_1, m_2$  collide elastically in  $d$  dimensions. Count the unknown outgoing velocity components and the conservation laws. With  $M = m_1 + m_2$ , the centre-of-mass velocity  $V$  and the relative velocity  $u = v_1 - v_2$ , show that the kinetic energy is  $\frac{1}{2}M|V|^2 + \frac{1}{2}\mu|u|^2$  with  $\mu = m_1m_2/M$ , and deduce that the possible outcomes are exactly the relative velocities  $u'$  with  $|u'| = |u|$ . What is this set for  $d = 1$ ,  $d = 2$  and  $d = 3$  when the particles have a relative velocity,  $u \neq 0$ , and what is it when  $u = 0$ ? (ii) Now  $N$  unit masses on a line meet at once. Show that fixed total momentum  $P$  and energy  $E$  confine the outgoing momenta to the set with  $\sum_i (p_i - P/N)^2 = 2E - P^2/N$  in

the plane  $\sum_i p_i = P$ , and that this contains every rearrangement of the incoming momenta. Show that it is a sphere of dimension  $N - 2$  and radius  $\sqrt{2E - P^2/N} > 0$  unless all incoming momenta are equal, when it is a single point. What is it for  $N = 2$  and  $N = 3$ ? (iii) The lecture's figure of three colliding particles has incoming momenta  $(1.6, 0.3, -1.2)$ , and prints the generic outcome as  $(-0.94, -0.14, 1.78)$ . Compute  $P$ ,  $2E = \sum p_i^2$  and the radius of the circle. Check that the printed outcome has the same  $P$  and, to the rounding of its last digit, the same  $\sum p_i^2$ , and that it is not a rearrangement of the incoming momenta.

**Exercise 4 (the FPUT spring and the Toda spring).** The lecture's tutorial matches the Toda spring  $V_T(r) = \frac{a}{b} e^{-br} + ar - \frac{a}{b}$  to the FPUT spring  $\kappa(\frac{1}{2}r^2 + \frac{\alpha}{3}r^3)$  through cubic order, with  $b = -2\alpha$  and  $a = -\kappa/(2\alpha)$ , for  $\alpha \neq 0$  and  $\kappa > 0$ . (i) Expand  $V_T$  through order  $r^4$ . (ii) Show that with the matched constants  $V_T(r) - \kappa(\frac{1}{2}r^2 + \frac{\alpha}{3}r^3) = \frac{1}{6}\kappa\alpha^2r^4 + O(r^5)$ . (iii) Show that the matched Toda spring satisfies  $V_T(r) \geq 0$  for every  $r$ , with equality only at  $r = 0$ . (iv) Take  $\kappa = 1$ . Show that the FPUT spring has, besides its minimum at  $r = 0$ , a local maximum at  $r = -1/\alpha$ , of height  $1/(6\alpha^2)$ , and that it decreases without bound beyond it. For the lecture's  $\alpha = \frac{1}{4}$ , where is the barrier and how high is it? What happens to a single FPUT spring given more energy than that?

**Exercise 5 (counting conserved quantities in quantum mechanics).** Let  $H$  be a Hermitian  $n \times n$  matrix with distinct eigenvalues  $\lambda_1, \dots, \lambda_n$ , and  $P_k$  the orthogonal projector onto the eigenline of  $\lambda_k$ . (i) Show that  $HP_k = P_kH$ , that  $P_jP_k = \delta_{jk}P_k$ , and that  $P_1, \dots, P_n$  are linearly independent: every such  $H$  has  $n$  commuting, linearly independent conserved quantities. (ii) Show that  $P_1, \dots, P_n$  span the same space of matrices as  $1, H, \dots, H^{n-1}$ . (iii) For  $H = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}$  with  $a, b$  real and  $b \neq 0$ , find  $P_{\pm}$  explicitly. (iv) What do (i) and (ii) say about defining a quantum integrable system as one with as many commuting conserved quantities as the dimension, or as degrees of freedom?

**Exercise 6 ★.** *The Toda chain keeps the eigenvalues of a matrix.* The open Toda chain of  $N$  unit masses has

$$H = \frac{1}{2} \sum_{n=1}^N p_n^2 + \sum_{n=1}^{N-1} e^{q_n - q_{n+1}},$$

so  $\dot{q}_n = p_n$  and  $\dot{p}_n = e^{q_{n-1} - q_n} - e^{q_n - q_{n+1}}$ , the terms with  $q_0$  and  $q_{N+1}$  absent. Its Lax matrix, which Lecture 8 builds, is the symmetric tridiagonal matrix  $L$  with  $p_1, \dots, p_N$  on the diagonal and  $a_n = e^{(q_n - q_{n+1})/2}$  beside it. (i) For  $N = 2$ , show directly from Hamilton's equations that  $\text{tr } L$  and  $\det L$  are conserved, and conclude that the eigenvalues of  $L$  are. (ii) Two Toda particles that collide separate again:  $q_2 - q_1 \rightarrow +\infty$  as  $t \rightarrow \pm\infty$ , and their momenta tend to limits (take this as given: Lecture 12 proves it for two particles by solving the motion, and cites it for  $N$ ). Deduce that the outgoing momenta are the incoming ones, exchanged. (iii) For  $N = 3$ , show that  $\text{tr } L^3 = \sum_n p_n^3 + 3 \sum_{n=1}^2 e^{q_n - q_{n+1}}(p_n + p_{n+1})$  and that it is conserved. Why does it follow, with  $\text{tr } L = \sum_n p_n$  and  $\text{tr } L^2 = 2H$ , that the eigenvalues of  $L$  are conserved?